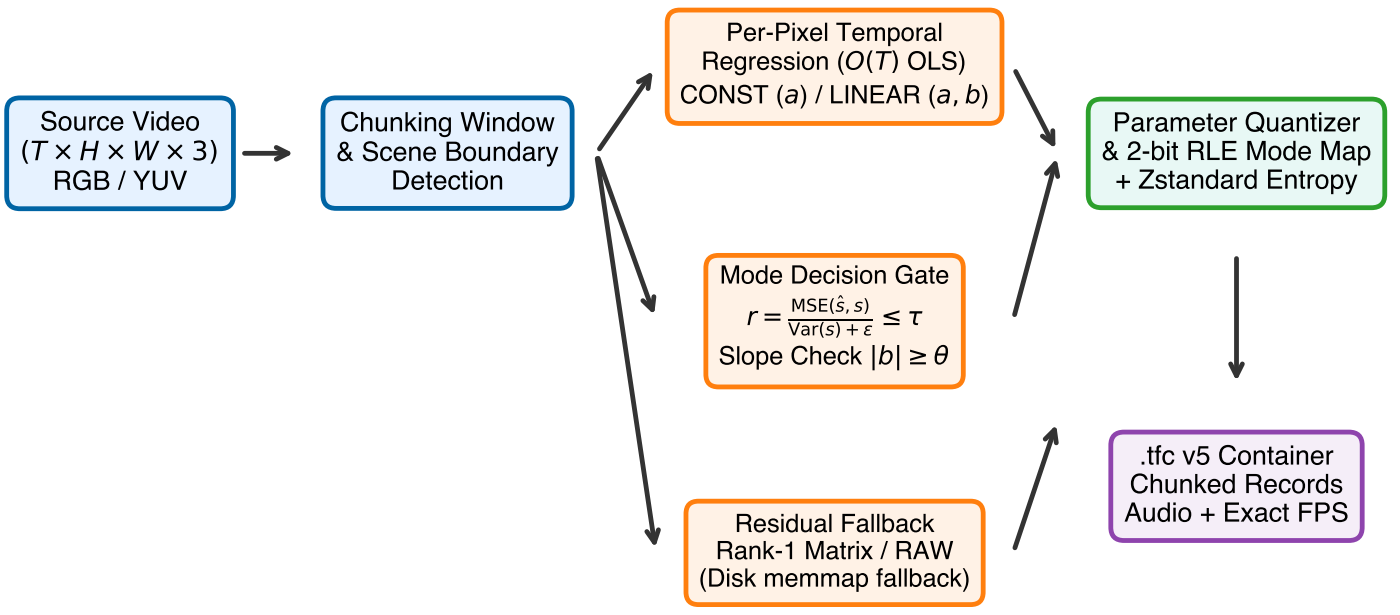
TimeFlowCodec v5 Architecture: Per-Pixel Temporal Streaming Pipeline
Source Video
$(T \times H \times W \times 3)$
RGB / YUV
Chunking Window
& Scene Boundary
Detection
Per-Pixel Temporal
Regression ($O(T)$ OLS)
CONST (a) / LINEAR (a, b)
Mode Decision Gate
$r = \frac{\mathrm{MSE}(\hat{s}, s)}{\mathrm{Var}(s) + \varepsilon} \le \tau$
Slope Check $|b| \ge \theta$
Residual Fallback
Rank-1 Matrix / RAW
(Disk memmap fallback)
Parameter Quantizer
& 2-bit RLE Mode Map
+ Zstandard Entropy
.tfc v5 Container
Chunked Records
Audio + Exact FPS